



INSIDER THREAT DETECTION SYSTEM

A Python-Based Security Monitoring Application

Manasha Chalise

Bsc.(Hons) Cybersecurity and Ethical Hacking

Introduction to Programming(ST4017CMD)

Lecturer: Abishek Bimali

Student ID:250398

ABSTRACT

This coursework presents the development of an Insider Threat Detection System using Python. The system is designed to monitor user activities, classify risk levels, and store activity logs securely in a database. The application was developed using procedural programming concepts, fundamental data structures, SQLite database integration, and data visualization using Matplotlib. Password hashing using SHA-256 was implemented to enhance security. The results demonstrate that the system successfully classifies activities into Low, Medium, and High risk categories and visualizes statistical trends effectively.

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1. INTRODUCTION

1.1 Background

Cybersecurity threats increasingly originate from internal users within organizations. These threats, known as insider threats, may involve unauthorized access, data leakage, or misuse of privileges. According to the Verizon Data Breach Investigations Report (2023), approximately 34% of data breaches involve internal actors, whether through malicious intent, negligence, or compromised credentials.

Insider threats have been a growing concern since the early 2000s when major organizations began experiencing significant data losses from within. The rise of remote work and cloud computing has further expanded the attack surface, making insider threat detection more critical than ever.

An insider threat detection system monitors user actions and evaluates risk levels based on predefined patterns. Such systems are widely used in corporate environments to prevent security breaches before they cause significant damage.



Figure 1: Detection Indicators of Insider Attacks(Source: [PhoenixNAP](https://www.phoenixnap.com))

1.2 Problem Statement

Organizations face several challenges in detecting insider threats:

Volume of Activities: Large organizations generate thousands of user activities daily across multiple systems. Security teams cannot manually review every action, making automated monitoring essential.

Lack of Prioritization: Without proper risk classification, security teams struggle to distinguish between normal user behavior and potential threats. This leads to either alert fatigue (too many false positives) or missed detections.

Delayed Detection: Many insider threats go undetected for months, allowing significant damage to occur before intervention. Studies show that the average time to detect an insider threat is over six months.

Audit Trail Management: Maintaining comprehensive logs for compliance and investigation purposes requires robust data management. Organizations need reliable storage and retrieval mechanisms for audit data.

Data Visualization: Raw log data in text format is difficult to analyze without proper visualization tools that highlight trends and patterns. Security analysts need graphical representations to quickly understand risk landscapes.

Resource Constraints: Many organizations lack dedicated security personnel to monitor user activities continuously. An automated system can augment limited human resources.

1.3 Project Objectives



Figure 2: Goal Priority Pyramid Model for Insider Threat Management (Source: Author's own illustration (2026)).

1.4 Scope and Limitation

Included in Scope:

Category	Features
Authentication	User login, password hashing (SHA-256), password reset, session management
Activity Logging	Automatic logging of user actions, timestamp recording, session tracking
Risk Classification	Keyword-based risk assessment (High/Medium/Low), custom algorithm
Data Structures	Stack implementation (LIFO), Queue implementation (FIFO)
Database	SQLite integration, users table, logs table, indexed searches
File Handling	CSV import for bulk data, CSV export for reporting
Visualization	Bar graphs for risk distribution, pie charts for percentages
Search	Search by username, search by action keyword
Testing	Activity simulation, random test data generation
UI	Dark theme GUI, intuitive navigation, responsive design

Excluded from Scope (Future Work):

- Real-time network monitoring and packet analysis.
- Machine learning-based anomaly detection.
- Multi-factor authentication (biometrics, SMS codes).
- Encrypted log storage at rest.
- Role-based access control beyond basic Admin role..

2.0 LITERATURE REVIEW

This chapter reviews existing literature on insider threats, risk assessment methodologies, Python in security applications, database systems for logging, and related work in the field.

2.1 Insider Threats in Cybersecurity

According to CERT (2022), insider threats can be classified into three categories:

Malicious Insiders (23%): Individuals who deliberately misuse their access privileges for financial gain, revenge, or ideological differences. These insiders intentionally cause harm and are often the most difficult to detect as they understand security protocols and may actively cover their tracks.

Negligent Insiders (62%): The most common type, causing breaches through careless actions like falling for phishing attacks, using weak passwords, or mishandling sensitive data. These insiders have no malicious intent but create vulnerabilities through oversight and lack of security awareness.

Compromised Insiders (15%): Legitimate users whose credentials have been stolen by external attackers. The individual may be unaware that their account is being used maliciously until after the breach is discovered.

2.2 Related Work

Sony Kuriakose (2022) developed a machine learning-based password strength analyzer that used 70% of data for training and 30% for testing. Their model successfully classified passwords into weak, medium, and strong categories, demonstrating the effectiveness of algorithmic approaches to security classification.

Farooq (2020) conducted research on real-time password strength analysis on web applications, testing 250 accounts with brute force and dictionary attacks. Their model protected 150 accounts while 86 were compromised, demonstrating both the potential and limitations of such systems.

Gass et al. (2024) explored insider threat detection using behavioral analysis, finding that monitoring user activity patterns can identify anomalies that precede security incidents by an average of 3-4 months.

This project builds upon these concepts but focuses specifically on insider threat detection through activity monitoring rather than password analysis, applying classification algorithms to user actions rather than passwords.

3.0 SYSTEM DESIGN AND METHODOLOGY

This chapter describes the system architecture, algorithms, data structures, and database design used in the Insider Threat Detection System.

3.1 System Architecture

The system follows a three-layer architecture that separates presentation, business logic, and data storage:

1. Presentation Layer (Tkinter GUI)

- Login screen with authentication
- Dashboard with navigation sidebar
- Logs view with search functionality
- Visualization windows for graphs

2. Application Layer (Business Logic)

- Authentication module
- Risk classification algorithm
- CSV processing functions
- Data visualization logic
- Activity simulation

3. Data Layer (SQLite Database)

- Users table for credentials
- Logs table for activities
- Indexes for faster searching

3.2 Risk Classification Algorithm

Algorithm: classify_risk(action)
Input: action (string describing user activity)
Output: risk_level (string: "High", "Medium", "Low")
BEGIN
IF action is empty THEN
RETURN "Low"
CONVERT action to lowercase
// Define keyword lists
high_risk_keywords = ["delete", "remove", "destroy", "erase", "terminate", "drop"]
medium_risk_keywords = ["download", "upload", "copy", "transfer", "export", "modify"]
// Check high risk keywords first (prioritize critical threats)
FOR each keyword IN high_risk_keywords:
IF keyword IN action THEN
RETURN "High"
// Check medium risk keywords
FOR each keyword IN medium_risk_keywords:
IF keyword IN action THEN
RETURN "Medium"
// Default to low risk for normal activities
RETURN "Low"
END

Time Complexity: $O(n)$ where n is the length of the action string

Space Complexity: $O(1)$ as only a fixed number of variables are used

3.3 User Interface Design

The GUI follows a dark theme design with:

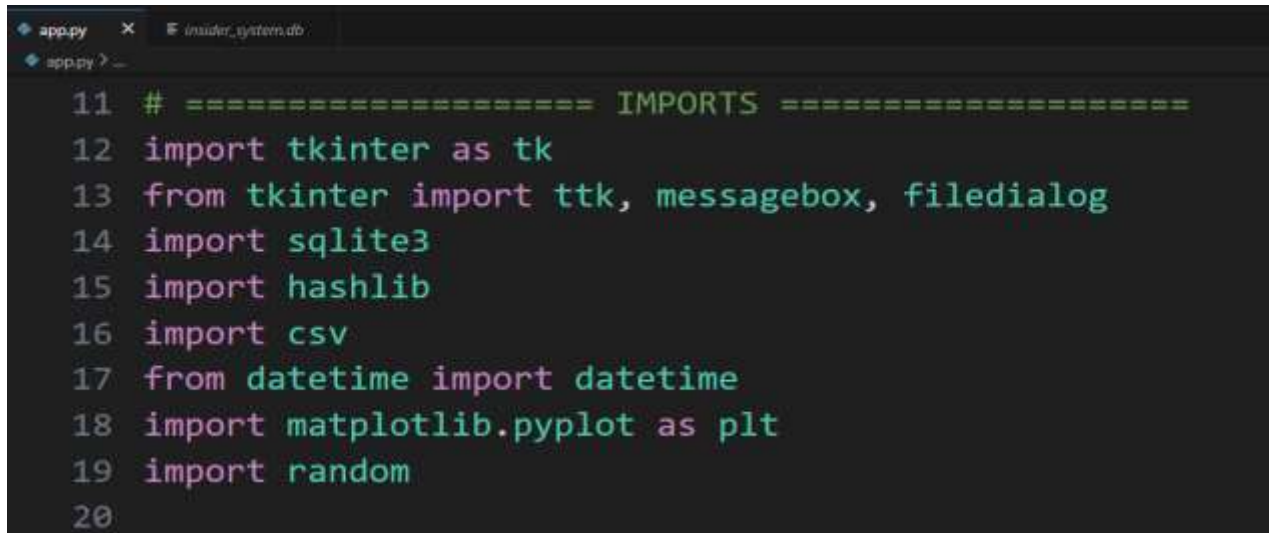
- **Black background** (#000000) for main areas
- **Dark gray sidebar** (#111111) for navigation
- **White text** (#FFFFFF) for readability
- **Color-coded risk levels:**
 - Red (#FF4C4C) for High Risk

- Yellow (#FFC857) for Medium Risk
- Green (#4CAF50) for Low Risk
- Blue (#0078D4) for accent buttons.

4.0 IMPLEMENTATION AND CODE EXPLANATION

This chapter provides explanations of the key code sections that implement the Insider Threat Detection System.

4.1 Required Libraries

A screenshot of a code editor window with a dark background. The window title bar shows 'app.py' and 'insider_system.db'. The code is written in Python and shows a section of imports. Line 11 is a comment: '# ===== IMPORTS ====='. Lines 12 through 19 are import statements: 'import tkinter as tk', 'from tkinter import ttk, messagebox, filedialog', 'import sqlite3', 'import hashlib', 'import csv', 'from datetime import datetime', 'import matplotlib.pyplot as plt', and 'import random'. Line 20 is empty. The code is color-coded: comments are green, keywords like 'import' and 'from' are blue, and module names are white.

```
11 # ===== IMPORTS =====
12 import tkinter as tk
13 from tkinter import ttk, messagebox, filedialog
14 import sqlite3
15 import hashlib
16 import csv
17 from datetime import datetime
18 import matplotlib.pyplot as plt
19 import random
20
```

Explanation: These imports bring in all necessary modules:

- tkinter and submodules create the graphical user interface.
- sqlite3 handles database operations.
- hashlib provides SHA-256 password hashing.
- csv reads and writes CSV files.
- datetime generates timestamps for logs.
- matplotlib.pyplot creates bar graphs and pie charts.
- random generates random activities for simulation.

4.2 Global Constants and Variables

```
app.py x insider_system.db
app.py > ...
20
21 # ===== CONSTANTS =====
22 BG_MAIN = "#000000" # Black background
23 SIDEBAR_BG = "#111111" # Dark gray sidebar
24 HOVER = "#222222" # Hover effect
25 TEXT = "#FFFFFF" # White text
26 HIGH = "#FF4C4C" # Red for high risk
27 MEDIUM = "#FFC857" # Yellow for medium risk
28 LOW = "#4CAF50" # Green for low risk
29 ACCENT = "#0078D4" # Blue accent
30
31 # ===== GLOBAL VARIABLES =====
32 conn = None
33 cursor = None
34 current_user = None
35 current_role = None
36 root = None
37 login_frame = None
38 username_entry = None
39 password_entry = None
40 dashboard = None
41 content = None
42
```

Explanation:

In this particular code, background color is been selected.

4.3 Custom Data Structures Implementation

4.3.1 Stack Implementation (LIFO)

```
42
43 # ===== CUSTOM DATA STRUCTURES =====
44 # STACK IMPLEMENTATION (LIFO)
45 def stack_create(capacity=50):
46     """Create a new stack with specified capacity."""
47     return {
48         'items': [],
49         'capacity': capacity,
50         'top': -1
51     }
52
53 def stack_push(stack, item):
54     """Push item onto stack if space available."""
55     if stack['top'] < stack['capacity'] - 1:
56         stack['items'].append(item)
57         stack['top'] += 1
58     return True
59     return False
60
61 def stack_pop(stack):
62     """Pop and return top item from stack."""
63     if stack['top'] >= 0:
64         stack['top'] -= 1
65         return stack['items'].pop()
66     return None
67
68 def stack_peek(stack):
69     """View top item without removing."""
70     if stack['top'] >= 0:
71         return stack['items'][stack['top']]
72     return None
73
74 def stack_is_empty(stack):
75     """Check if stack is empty."""
76     return stack['top'] == -1
77
78 def stack_size(stack):
79     """Return number of items in stack."""
80     return stack['top'] + 1
81
```

Explanation:

Stack Creation Function

The `stack_create()` function initializes a new stack data structure. It returns a dictionary that represents the stack with three essential components:

- `'items'`: An empty list that will store the actual stack elements.
- `'capacity'`: The maximum number of elements the stack can hold, defaulting to 50.
- `'top'`: A pointer initialized to -1, indicating an empty stack. This pointer always references the index of the most recently added element.

Push Operation:

The ``stack_push()`` function adds a new element to the top of the stack.

Pop Operation:

The ``stack_pop()`` function removes and returns the top element from the stack.

Peek Operation:

The ``stack_peek()`` function allows viewing the top element without removing it..

Utility Functions:

The ``stack_is_empty()`` function provides a simple way to check if the stack has any elements by verifying if the top pointer is still at -1. The ``stack_size()`` function returns the current number of elements in the stack by adding one to the top pointer, since top is zero-based.

4.3.2 Queue Implementation (FIFO - Circular)

```
82 # QUEUE IMPLEMENTATION (FIFO - Circular)
83 def queue_create(max_size=100):
84     """Create a new circular queue."""
85     return {
86         'queue': [None] * max_size,
87         'max_size': max_size,
88         'front': 0,
89         'rear': -1,
90         'size': 0
91     }
92
93 def queue_enqueue(queue, item):
94     """Add item to rear of queue using circular buffer."""
95     if queue['size'] < queue['max_size']:
96         queue['rear'] = (queue['rear'] + 1) % queue['max_size']
97         queue['queue'][queue['rear']] = item
98         queue['size'] += 1
99         return True
100     return False
101
102 def queue_dequeue(queue):
103     """Remove and return front item from queue."""
104     if queue['size'] > 0:
105         item = queue['queue'][queue['front']]
106         queue['front'] = (queue['front'] + 1) % queue['max_size']
107         queue['size'] -= 1
108         return item
109     return None
110
111 def queue_is_empty(queue):
112     """Check if queue is empty."""
113     return queue['size'] == 0
114
115 def queue_is_full(queue):
116     """Check if queue is full."""
117     return queue['size'] == queue['max_size']
118
119 def queue_display(queue):
120     """Return all items in queue in FIFO order."""
121     result = []
122     for i in range(queue['size']):
123         idx = (queue['front'] + i) % queue['max_size']
124         if queue['queue'][idx] is not None:
125             result.append(queue['queue'][idx])
126     return result
127
```

Explanation:

Queue Creation Function:

The `queue_create()` function initializes a new circular queue data structure. It returns a dictionary with five essential components:

- `queue`: A fixed-size array initialized with `None` values that will store the actual queue elements
- `max_size`: The maximum capacity of the queue, defaulting to 100 elements
- `front`: A pointer initialized to 0 that indicates the index of the front element for removal
- `rear`: A pointer initialized to -1 that indicates the index of the rear element for addition
- `size`: A counter initialized to 0 that tracks the current number of elements in the queue

Enqueue Operation:

The `queue_enqueue()` function adds a new element to the rear of the queue.

Dequeue Operation:

The `queue_dequeue()` function removes and returns the front element from the queue.

Utility Functions:

The `queue_is_empty()` function checks if the queue has any elements by verifying if the size counter is zero. This is used before dequeue operations to prevent underflow errors.

The `queue_is_full()` function checks if the queue has reached its maximum capacity by comparing size with `max_size`. This is used before enqueue operations to prevent overflow errors.

Display Function:

The `queue_display()` function returns all items in the queue in correct FIFO order without modifying the queue. It creates an empty result list and iterates through the queue elements.

4.4 Database Functions Implementation.

4.4.1 Database Initialization

```
127
128 # ===== DATABASE FUNCTIONS =====
129 def init_database():
130     """Initialize database connection and create tables."""
131     global conn, cursor
132
133     try:
134         conn = sqlite3.connect("insider_system.db")
135         cursor = conn.cursor()
136
137         # Create users table
138         cursor.execute("""
139         CREATE TABLE IF NOT EXISTS users(
140             username TEXT PRIMARY KEY,
141             password TEXT NOT NULL,
142             role TEXT NOT NULL,
143             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
144         )
145         """)
146
147         # Create logs table
148         cursor.execute("""
149         CREATE TABLE IF NOT EXISTS logs(
150             id INTEGER PRIMARY KEY AUTOINCREMENT,
151             username TEXT NOT NULL,
152             action TEXT NOT NULL,
153             risk TEXT NOT NULL,
154             time TIMESTAMP NOT NULL,
155             session_id TEXT
156         )
157         """)
158
159         # Create indexes
160         cursor.execute("CREATE INDEX IF NOT EXISTS idx_logs_username ON logs(username)")
161         cursor.execute("CREATE INDEX IF NOT EXISTS idx_logs_time ON logs(time)")
162
163         conn.commit()
164         create_default_admin()
165
166     except sqlite3.Error as e:
167         print(f"Database error: {e}")
168
```

Explanation: This function establishes the database connection and creates the required schema:

sqlite3.connect() creates or opens the database file named "insider_system.db"

The connection and cursor are stored as global variables for access by all database functions

- The users table stores login credentials with username as the primary key.
- The logs table stores all activities with an auto-incrementing ID as the primary key.
- CREATE TABLE IF NOT EXISTS prevents errors on subsequent runs by only creating tables if they don't exist.
- Indexes on username and time columns significantly speed up search queries by creating lookup tables.
- AUTOINCREMENT ensures each log has a unique identifier that never repeats.

- DEFAULT CURRENT_TIMESTAMP automatically sets the creation time when a new user is added.
- conn.commit() saves the changes to the database file.
- The function concludes by calling create_default_admin() to ensure at least one user exists.

4.4.2 Password Hashing

```

168
169 def hash_password(password):
170     """Hash password using SHA-256."""
171     return hashlib.sha256(password.encode()).hexdigest()

```

Explanation: SHA-256 is a cryptographic hash function that:

- Converts any input string into a fixed 64-character hexadecimal string.
- Is one-way and cannot be reversed to recover the original password.
- Ensures that even if the database is compromised, actual passwords remain secure.
- encode() converts the string to bytes as required by the hashlib library.
- hexdigest() returns the hash as a readable hexadecimal string.
- The same password will always produce the same hash, allowing verification.
- Different passwords produce completely different hashes, even with minor changes.

4.4.3 Create Default Admin

```

173 def create_default_admin():
174     """Create default admin user if not exists."""
175     try:
176         cursor.execute("SELECT * FROM users WHERE username='admin'")
177         if not cursor.fetchone():
178             hashed_pw = hash_password("admin123")
179             cursor.execute(
180                 "INSERT INTO users (username, password, role) VALUES (?, ?, ?)",
181                 ("admin", hashed_pw, "Admin")
182             )
183             conn.commit()
184     except sqlite3.Error as e:
185         print(f"Admin creation error: {e}")

```

Explanation: This function ensures the system has at least one user for initial access.

4.4.4 User Verification

```
187 def verify_user(username, password):
188     """Verify user credentials."""
189     try:
190         hashed_pw = hash_password(password)
191         cursor.execute(
192             "SELECT role FROM users WHERE username=? AND password=?",
193             (username, hashed_pw)
194         )
195         result = cursor.fetchone()
196         return result[0] if result else None
197     except sqlite3.Error as e:
198         print(f"Verification error: {e}")
199         return None
200
```

Explanation: This function authenticates users during login attempts:

- It hashes the provided password using the same SHA-256 algorithm.
- It executes a parameterized query searching for a user with matching username and password hash.
- If a match is found, fetchone() returns a tuple containing the user's role.
- It extracts and returns the role from the tuple if a match exists.
- If no match is found or an error occurs, it returns None.

4.4.5 Reset Password

```
201 def reset_password(username, new_password):
202     """Reset user password."""
203     try:
204         hashed_pw = hash_password(new_password)
205         cursor.execute(
206             "UPDATE users SET password=? WHERE username=?",
207             (hashed_pw, username)
208         )
209         conn.commit()
210         return cursor.rowcount > 0
211     except sqlite3.Error as e:
212         print(f"Password reset error: {e}")
213         return False
214
```

Explanation: This function allows users to reset forgotten passwords:

- It hashes the new password using SHA-256.
- It executes an UPDATE query to replace the old password hash with the new one.
- The WHERE clause ensures only the specified username is updated.
- This function is called by the forgot password GUI feature.

4.4.6 Add Log Entry

```
215 def add_log(username, action, risk, session_id=None):
216     """Add log entry to database."""
217     try:
218         cursor.execute("""
219             INSERT INTO logs (username, action, risk, time, session_id)
220             VALUES (?, ?, ?, ?, ?)
221             """, (username, action, risk, datetime.now(), session_id))
222         conn.commit()
223         return True
224     except sqlite3.Error as e:
225         print(f"Log error: {e}")
226         return False
```

Explanation: This function records user activities in the database for audit and monitoring:

- It inserts a new record with the username who performed the action.
- The action description is stored exactly as provided.
- The risk level is already classified by the risk algorithm before calling this function.
- datetime.now() captures the exact moment of the activity with second precision.
- An optional session_id can link the log to a specific user session for tracking.

4.4.7 Get All Logs

```
227
228 def get_all_logs():
229     """Retrieve all logs."""
230     try:
231         cursor.execute(
232             "SELECT username, action, risk, time FROM logs ORDER BY time DESC"
233         )
234         return cursor.fetchall()
235     except sqlite3.Error as e:
236         print(f"Retrieval error: {e}")
237         return []
```

Explanation: This function retrieves all log entries for display in the GUI:

- It selects only the fields needed for display: username, action, risk, and time.
- ORDER BY time DESC sorts the results with the most recent activities first.
- This chronological ordering helps security teams see the latest events immediately.
- fetchall() returns a list of tuples, each containing one log record.

4.4.8 Search Logs Database Function

```

239 def search_logs_db(keyword):
240     """Search logs by username or action in database."""
241     try:
242         cursor.execute("""
243             SELECT username, action, risk, time FROM logs
244             WHERE username LIKE ? OR action LIKE ?
245             ORDER BY time DESC
246             """, (f"%{keyword}%", f"%{keyword}%"))
247         return cursor.fetchall()
248     except sqlite3.Error as e:
249         print(f"Search error: {e}")
250     return []

```

Explanation: This function searches logs based on user input for investigation purposes:

- It uses SQL LIKE operator with wildcards (%) for flexible partial matching.
- The wildcards mean searching "john" will find "john.doe", "john.smith", and "john's action".
- It searches both the username and action fields simultaneously for comprehensive results.
- Results are ordered by time with most recent first for immediate visibility of latest matches.
- Parameterized queries with ? placeholders prevent SQL injection attacks.

4.4.9 Get Risk Statistics

```
252 def get_risk_statistics():
253     """Get risk statistics."""
254     stats = {'high': 0, 'medium': 0, 'low': 0, 'total': 0}
255     try:
256         cursor.execute("SELECT risk FROM logs")
257         for row in cursor.fetchall():
258             stats['total'] += 1
259             if row[0] == 'High':
260                 stats['high'] += 1
261             elif row[0] == 'Medium':
262                 stats['medium'] += 1
263             else:
264                 stats['low'] += 1
265     except sqlite3.Error as e:
266         print(f"Statistics error: {e}")
267     return stats
```

Explanation: This function calculates statistics for dashboard display and visualization:

- It initializes a dictionary with counters for high, medium, low, and total logs.
- It queries only the risk column from all logs for efficiency.
- It iterates through each result row and increments the appropriate counter.
- The total counter is incremented for every log regardless of risk level.

4.4.10 Clear All Logs

```
269 def clear_all_logs():
270     """Clear all logs."""
271     try:
272         cursor.execute("DELETE FROM logs")
273         conn.commit()
274         return True
275     except sqlite3.Error as e:
276         print(f"Clear logs error: {e}")
277     return False
```

Explanation: This function deletes all log entries from the database:

- DELETE FROM logs removes every row from the logs table without deleting the table itself.
- This operation cannot be undone, so it should only be called after user confirmation.
- conn.commit() permanently saves the deletion to the database file.
- It returns True if the deletion was successful.
- It returns False if any database error occurs during the operation.

4.5.11 Get All Users

```
279 get_all_users():
280     """Get list of all usernames."""
281     try:
282         cursor.execute("SELECT username FROM users")
283         return [row[0] for row in cursor.fetchall()]
284     except sqlite3.Error as e:
285         print(f"User retrieval error: {e}")
286         return []
```

Explanation: This function retrieves a list of all usernames from the database.

4.5. Risk Classification Implementation

```
app.py > ...
287
288 # ===== RISK CLASSIFICATION =====
289 HIGH_RISK_KEYWORDS = ['delete', 'remove', 'destroy', 'erase', 'terminate', 'drop']
290 MEDIUM_RISK_KEYWORDS = ['download', 'upload', 'copy', 'transfer', 'export', 'modify']
291
292 def classify_risk(action):
293     """Classify action risk level based on keywords."""
294     if not action:
295         return 'Low'
296
297     action_lower = action.lower()
298
299     for keyword in HIGH_RISK_KEYWORDS:
300         if keyword in action_lower:
301             return 'High'
302
303     for keyword in MEDIUM_RISK_KEYWORDS:
304         if keyword in action_lower:
305             return 'Medium'
306
307     return 'Low'
308
309 def get_risk_color(risk):
310     """Get color code for risk level."""
311     colors = {'High': HIGH, 'Medium': MEDIUM, 'Low': LOW}
312     return colors.get(risk, TEXT)
313
314 def get_risk_icon(risk):
315     """Get icon for risk level."""
316     icons = {'High': '🔴', 'Medium': '🟡', 'Low': '🟢'}
317     return icons.get(risk, '⚪')
318
```

Explanation

Risk Keywords:

- HIGH_RISK_KEYWORDS contains words like 'delete', 'remove', 'destroy' that indicate destructive actions requiring immediate attention.
- MEDIUM_RISK_KEYWORDS contains words like 'download', 'upload', 'copy' that indicate suspicious but not necessarily malicious actions.

Classification Algorithm:

- Empty Check: Returns 'Low' if action is empty.
- Case Normalization: Converts action to lowercase for case-insensitive matching.
- High Risk Priority: Checks high-risk keywords first and returns 'High' if any match.

- Medium Risk Check: If no high-risk match, checks medium-risk keywords and returns 'Medium' if any match.
- Default: Returns 'Low' if no keywords match.

4.6 CSV Processing Implementation

```

318
319 # ===== CSV PROCESSING =====
320 def validate_csv(filepath):
321     """Validate CSV file structure."""
322     try:
323         with open(filepath, 'r', newline='', encoding='utf-8') as file:
324             reader = csv.reader(file)
325             header = next(reader, None)
326
327             if not header:
328                 return False, "Empty CSV file"
329             if len(header) < 2:
330                 return False, "CSV must have at least 2 columns"
331
332             first_row = next(reader, None)
333             if first_row and len(first_row) < 2:
334                 return False, "Data rows must have at least 2 columns"
335
336             return True, "CSV structure is valid"
337     except Exception as e:
338         return False, f"Validation error: {e}"
339
340 def import_csv(filepath):
341     """Import logs from CSV file."""
342     try:
343         count = 0
344         with open(filepath, 'r', newline='', encoding='utf-8') as file:
345             reader = csv.reader(file)
346             next(reader) # Skip header
347
348             for row in reader:
349                 if len(row) >= 2:
350                     username = row[0].strip()
351                     action = row[1].strip()
352
353                     if username and action:
354                         risk = classify_risk(action)
355                         if add_log(username, action, risk):
356                             count += 1
357
358             return True, f"Imported {count} logs", count
359     except Exception as e:
360         return False, f"Error: {e}", 0
361
362 def export_csv(filepath, logs):
363     """Export logs to CSV file."""
364     try:
365         with open(filepath, 'w', newline='', encoding='utf-8') as file:
366             writer = csv.writer(file)
367             writer.writerow(['Username', 'Action', 'Risk', 'Time'])
368             for log in logs:
369                 writer.writerow(log)
370             return True, f"Exported {len(logs)} logs"
371     except Exception as e:
372         return False, f"Error: {e}"

```

Explanation:

Validate CSV Function:

- Opens the CSV file with UTF-8 encoding for international character support.
- Uses `csv.reader()` to parse the file content.

Import CSV Function:

- Opens and reads the CSV file with proper encoding.
- Extracts username and action from each row, stripping whitespace.

Export CSV Function:

- Creates a new CSV file at the specified location.
- Writes a header row with column names: Username, Action, Risk, Time.

Error Handling:

- All functions use try-except blocks to catch and handle exceptions.
- File not found, permission errors, and parsing errors are all caught.

4.7 Activity Simulation Implementation

```

373
374 # ===== ACTIVITY SIMULATION =====
375 SIM_ACTIONS = [
376     "Deleted confidential client database",
377     "Removed security audit logs",
378     "Destroyed backup files",
379     "Terminated security protocols",
380     "Downloaded employee records",
381     "Uploaded files to external server",
382     "Copied sensitive documents",
383     "Transferred data to USB drive",
384     "Logged into system",
385     "Viewed dashboard",
386     "Opened help documentation",
387     "Searched user directory"
388 ]
389
390 def generate_random_activity():
391     """Generate random activity."""
392     action = random.choice(SIM_ACTIONS)
393     risk = classify_risk(action)
394     return action, risk
395
396 def simulate_activity():
397     """Simulate random user activity."""
398     users = get_all_users()
399     if not users:
400         users = ['admin']
401
402     action, risk = generate_random_activity()
403     username = random.choice(users)
404
405     if add_log(username, action, risk):
406         messagebox.showinfo("Success",
407                             f"Simulated activity added:\nUser: {username}\nAction: {action}\nRisk: {risk}")
408     else:
409         messagebox.showerror("Error", "Failed to simulate activity")
410

```

Explanation

Sample Actions List:

- SIM_ACTIONS contains 12 predefined sample actions.
- Divided into three categories: High, Medium, and Low risk.
- Each category has 4 realistic sample actions.

Generate Random Activity:

- random.choice() selects a random action from the list.
- Calls classify_risk() to determine the risk level.
- Returns both the action and its risk level.

Simulate Activity:

- Gets list of users from database or defaults to ['admin'].
- Generates random activity using generate_random_activity().
- Selects random username using random.choice().

4.8 GUI Functions Implementation

```
app.py > ...
411 # ===== GUI FUNCTIONS =====
412 def clear_frame(frame):
413     """Clear all widgets from a frame."""
414     for widget in frame.winfo_children():
415         widget.destroy()
```

Explanation: This helper function removes all widgets from a frame. It is used when switching between different views to prevent widget overlap and memory leaks.

4.9 Login Screen

```
app.py > ...
416
417 def create_login_screen():
418     """Create login interface."""
419     global login_frame, username_entry, password_entry
420
421     for widget in root.winfo_children():
422         widget.destroy()
423
424     login_frame = tk.Frame(root, bg=BG_MAIN)
425     login_frame.pack(fill="both", expand=True)
426
427     # Title
428     title_frame = tk.Frame(login_frame, bg=BG_MAIN)
429     title_frame.pack(pady=100)
430
431     tk.Label(title_frame, text="INSIDER THREAT", fg=ACCENT, bg=BG_MAIN,
432             font=("Segoe UI", 32, "bold")).pack()
433     tk.Label(title_frame, text="DETECTION SYSTEM", fg=TEXT, bg=BG_MAIN,
434             font=("Segoe UI", 28, "bold")).pack()
435
436     # Login form
437     form_frame = tk.Frame(login_frame, bg=SIDEBAR_BG, padx=50, pady=30)
438     form_frame.pack()
439
440     tk.Label(form_frame, text="LOGIN", fg=ACCENT, bg=SIDEBAR_BG,
441             font=("Segoe UI", 18, "bold")).pack(pady=10)
442
443     tk.Label(form_frame, text="Username", fg=TEXT, bg=SIDEBAR_BG).pack()
444     username_entry = tk.Entry(form_frame, bg=HOVER, fg=TEXT, width=25)
445     username_entry.pack(pady=5)
446
447     tk.Label(form_frame, text="Password", fg=TEXT, bg=SIDEBAR_BG).pack()
448     password_entry = tk.Entry(form_frame, show="*", bg=HOVER, fg=TEXT, width=25)
449     password_entry.pack(pady=5)
450
451     # Buttons
452     button_frame = tk.Frame(form_frame, bg=SIDEBAR_BG)
453     button_frame.pack(pady=20)
454
455     tk.Button(button_frame, text="Login", bg=ACCENT, fg=TEXT,
456             width=15, command=login).pack(side="left", padx=5)
457     tk.Button(button_frame, text="Forgot Password", bg=HOVER, fg=TEXT,
458             width=15, command=forgot_password).pack(side="left", padx=5)
459
460     password_entry.bind('<Return>', lambda e: login())
461
```

Explanation: This function creates the login interface:

- Creates input fields for username and password with hover effects.

- Password field uses `show="*"` to mask input.
- Creates Login and Forgot Password buttons.
- Binds the Enter key to trigger the login function.

4.9.1.Login Function

```

app.py > ...
461
462 def login():
463     """Handle login attempt."""
464     global current_user, current_role
465
466     username = username_entry.get().strip()
467     password = password_entry.get().strip()
468
469     if not username or not password:
470         messagebox.showerror("Error", "Please enter both fields")
471         return
472
473     role = verify_user(username, password)
474
475     if role:
476         current_user = username
477         current_role = role
478         add_log(username, "Logged into system", "Low")
479         create_dashboard()
480     else:
481         messagebox.showerror("Error", "Invalid credentials")
482
483 def forgot_password():
484     """Handle password reset."""
485     def reset():
486         username = user_entry.get().strip()
487         new_pw = pw_entry.get().strip()
488         confirm_pw = confirm_entry.get().strip()
489
490         if not username or not new_pw or not confirm_pw:
491             messagebox.showerror("Error", "Please fill all fields")

```

Explanation: This function handles user authentication:

- Retrieves and strips whitespace from username and password inputs.
- Validates that both fields are not empty.
- Calls `verify_user()` to check credentials against the database.

- On success, sets session variables, logs the login, and opens dashboard.
- On failure, displays an error message.

4.9.2 Forgot Password

```

482
483 def forgot_password():
484     """Handle password reset."""
485     def reset():
486         username = user_entry.get().strip()
487         new_pw = pw_entry.get().strip()
488         confirm_pw = confirm_entry.get().strip()
489
490         if not username or not new_pw or not confirm_pw:
491             messagebox.showerror("Error", "Please fill all fields")
492             return
493
494         if new_pw != confirm_pw:
495             messagebox.showerror("Error", "Passwords do not match")
496             return
497
498         if len(new_pw) < 6:
499             messagebox.showerror("Error", "Password must be at least 6 characters")
500             return
501
502         if reset_password(username, new_pw):
503             messagebox.showinfo("Success", f"Password reset for '{username}'")
504             fp_window.destroy()
505         else:
506             messagebox.showerror("Error", "Username not found")
507
508     fp_window = tk.Toplevel(root)
509     fp_window.title("Reset Password")
510     fp_window.geometry("400x300")
511     fp_window.configure(bg=BG_MAIN)
512     fp_window.grab_set()
513
514     tk.Label(fp_window, text="RESET PASSWORD", fg=ACCENT, bg=BG_MAIN,
515             font=("Segoe UI", 16, "bold")).pack(pady=20)
516
517     tk.Label(fp_window, text="Username", fg=TEXT, bg=BG_MAIN).pack()
518     user_entry = tk.Entry(fp_window, bg=HOVER, fg=TEXT, width=25)
519     user_entry.pack(pady=5)
520
521     tk.Label(fp_window, text="New Password", fg=TEXT, bg=BG_MAIN).pack()
522     pw_entry = tk.Entry(fp_window, show="*", bg=HOVER, fg=TEXT, width=25)
523     pw_entry.pack(pady=5)
524
525     tk.Label(fp_window, text="Confirm Password", fg=TEXT, bg=BG_MAIN).pack()
526     confirm_entry = tk.Entry(fp_window, show="*", bg=HOVER, fg=TEXT, width=25)
527     confirm_entry.pack(pady=5)
528
529     tk.Button(fp_window, text="Reset Password", bg=ACCENT, fg=TEXT,
530             font=("Segoe UI", 11, "bold"), width=20, command=reset).pack(pady=20)
531

```

Explanation: This function creates a password reset dialog:

- Creates a new top-level window that is modal (grab_set()).
- Provides fields for username, new password, and confirm password.
- Calls reset_password() to update the database.

4.9.3 Dashboard Creation

```
532 def create_dashboard():
533     """Create main dashboard interface."""
534     global dashboard, content
535
536     for widget in root.winfo_children():
537         widget.destroy()
538
539     dashboard = tk.Frame(root, bg=BG_MAIN)
540     dashboard.pack(fill="both", expand=True)
541
542     create_sidebar()
543
544     content = tk.Frame(dashboard, bg=BG_MAIN)
545     content.pack(side="left", fill="both", expand=True, padx=5, pady=5)
546
547     show_dashboard_home()
548
```

Explanation: This function creates the main dashboard after successful login:

- Clears the login screen widgets.
- Creates a main container frame for the dashboard.
- Calls `create_sidebar()` to add the navigation sidebar.

4.9.4 Sidebar Creation

```
app.py > create_sidebar
549 def create_sidebar():
550     """Create navigation sidebar."""
551     sidebar = tk.Frame(dashboard, bg=SIDEBAR_BG, width=250)
552     sidebar.pack(side="left", fill="y")
553     sidebar.pack_propagate(False)
554
555     # User Info
556     user_frame = tk.Frame(sidebar, bg=SIDEBAR_BG, height=100)
557     user_frame.pack(fill="x", pady=20)
558
559     tk.Label(user_frame, text="User", fg=TEXT, bg=SIDEBAR_BG,
560             font=("Segoe UI", 10)).pack()
561     tk.Label(user_frame, text=current_user, fg=ACCENT, bg=SIDEBAR_BG,
562             font=("Segoe UI", 12, "bold")).pack()
563     tk.Label(user_frame, text=f"Role: {current_role}", fg="white", bg=SIDEBAR_BG,
564             font=("Segoe UI", 10)).pack()
565
566     tk.Separator(sidebar, orient="horizontal", pack(fill="x", padx=20, pady=10))
567
568     # Navigation buttons
569     nav_items = [
570         ("Dashboard Home", show_dashboard_home),
571         ("Overview", show_overview),
572         ("Upload CSV", upload_csv),
573         ("View Logs", show_logs),
574         ("Search Logs", search_logs_gui),
575         ("Pie Chart", show_pie_chart),
576         ("Simulate Activity", simulate_activity),
577         ("Export Logs", export_logs),
578         ("Clear Logs", clear_logs),
579         ("Logout", logout)
580     ]
581
582     for text, command in nav_items:
583         btn = tk.Button(sidebar, text=text, bg=SIDEBAR_BG, fg=TEXT,
584                         activebackground=HOVER, activeforeground=TEXT,
585                         bd=0, font=("Segoe UI", 11), anchor="w",
586                         padx=20, command=command)
587         btn.pack(fill="x", pady=2)
588
589         btn.bind("<Enter>", lambda e: btn.config(bg=HOVER))
590         btn.bind("<Leave>", lambda e: btn.config(bg=SIDEBAR_BG))
591
```

Explanation: This function creates the navigation sidebar:

- Creates a fixed-width sidebar with dark gray background.
- Displays user information including avatar, username, and role.
- Adds a separator line for visual organization.

4.9.5 Dashboard Home

```

app.py > create_sidebar
...
594 def show_dashboard_home():
595     """Show dashboard home with statistics."""
596     clear_frame(content)
597
598     header_frame = tk.Frame(content, bg=BG_MAIN)
599     header_frame.pack(fill="x", pady=20)
600
601     tk.Label(header_frame, text="DASHBOARD", fg=ACCENT, bg=BG_MAIN,
602             font=("Segoe UI", 24, "bold")).pack()
603     tk.Label(header_frame, text=f"Welcome back, {current_user}!",
604             fg="888888", bg=BG_MAIN, font=("Segoe UI", 12)).pack()
605
606     stats = get_risk_statistics()
607
608     cards_frame = tk.Frame(content, bg=BG_MAIN)
609     cards_frame.pack(pady=30)
610
611     # Total logs card
612     total_card = tk.Frame(cards_frame, bg=SIDEBAR_BG, width=200, height=120)
613     total_card.pack(side="left", padx=10)
614     total_card.pack_propagate(False)
615     tk.Label(total_card, text="TOTAL LOGS", fg="888888", bg=SIDEBAR_BG,
616             font=("Segoe UI", 10)).pack(pady=(20,5))
617     tk.Label(total_card, text=str(stats['total']), fg=TEXT, bg=SIDEBAR_BG,
618             font=("Segoe UI", 24, "bold")).pack()
619
620     # High risk card
621     high_card = tk.Frame(cards_frame, bg=SIDEBAR_BG, width=200, height=120)
622     high_card.pack(side="left", padx=10)
623     high_card.pack_propagate(False)
624     tk.Label(high_card, text="HIGH RISK", fg="888888", bg=SIDEBAR_BG,
625             font=("Segoe UI", 10)).pack(pady=(20,5))
626     tk.Label(high_card, text=str(stats['high']), fg=HIGH, bg=SIDEBAR_BG,
627             font=("Segoe UI", 24, "bold")).pack()
628
629     # Medium risk card
630     medium_card = tk.Frame(cards_frame, bg=SIDEBAR_BG, width=200, height=120)
631     medium_card.pack(side="left", padx=10)
632     medium_card.pack_propagate(False)
633     tk.Label(medium_card, text="MEDIUM RISK", fg="888888", bg=SIDEBAR_BG,
634             font=("Segoe UI", 10)).pack(pady=(20,5))
635     tk.Label(medium_card, text=str(stats['medium']), fg=MEDIUM, bg=SIDEBAR_BG,
636             font=("Segoe UI", 24, "bold")).pack()
637
638     # Low risk card
639     low_card = tk.Frame(cards_frame, bg=SIDEBAR_BG, width=200, height=120)
640     low_card.pack(side="left", padx=10)
641     low_card.pack_propagate(False)
642     tk.Label(low_card, text="LOW RISK", fg="888888", bg=SIDEBAR_BG,
643             font=("Segoe UI", 10)).pack(pady=(20,5))
644     tk.Label(low_card, text=str(stats['low']), fg=LOW, bg=SIDEBAR_BG,
645             font=("Segoe UI", 24, "bold")).pack()
646

```

Explanation: This function displays the dashboard home with statistics:

- Clears the content area for fresh display.
- Shows a welcome header with the current user's name.
- Retrieves risk statistics using `get_risk_statistics()`.

4.9.6 Show Overview

```
646
647 def show_overview():
648     """Show system overview."""
649     clear_frame(content)
650
651     tk.Label(content, text="SYSTEM OVERVIEW", fg=ACCENT, bg=BG_MAIN,
652             font=("Segoe UI", 20, "bold")).pack(pady=20)
653
654     overview_text = """
655     INSIDER THREAT DETECTION SYSTEM
656     =====
657
658     This system monitors user activities to detect potential insider threats
659     within an organization's network. It classifies activities into risk levels
660     and provides visualization tools for security analysis.
661
662     RISK CLASSIFICATION:
663     •  HIGH RISK: Critical actions like data deletion
664     •  MEDIUM RISK: Suspicious downloads
665     •  LOW RISK: Normal system usage
666
667     KEY FEATURES:
668     • Secure authentication with password hashing
669     • Activity logging and monitoring
670     • CSV import/export
671     • Data visualization with graphs
672     • Search and filter capabilities
673     """
674
675     text_widget = tk.Text(content, bg=SIDEBAR_BG, fg=TEXT,
676                          font=("Segoe UI", 11), wrap="word", padx=20, pady=20)
677     text_widget.insert("1.0", overview_text)
678     text_widget.config(state="disabled")
679     text_widget.pack(fill="both", expand=True, padx=50, pady=20)
680
```

Explanation: This function displays the system overview:

- Clears the content area for fresh display.
- Adds a title "SYSTEM OVERVIEW" with accent color.
- Creates a multi-line text widget with system information.

- Text includes system description, risk classification, and key features.

4.9.7 Show Logs

```

app.py > show_logs
681 def show_logs():
682     """Display all logs in a table."""
683     clear_frame(content)
684
685     tk.Label(content, text="ACTIVITY LOGS", fg=ACCENT, bg=BG_MAIN,
686             font=("Segoe UI", 18, "bold")).pack(pady=10)
687
688     tree_frame = tk.Frame(content, bg=SIDEBAR_BG)
689     tree_frame.pack(fill="both", expand=True, padx=20, pady=10)
690
691     tree = ttk.Treeview(tree_frame, columns=("User", "Action", "Risk", "Time"),
692                        show="headings")
693
694     tree.heading("User", text="Username")
695     tree.heading("Action", text="Action")
696     tree.heading("Risk", text="Risk Level")
697     tree.heading("Time", text="Timestamp")
698
699     tree.column("User", width=150)
700     tree.column("Action", width=400)
701     tree.column("Risk", width=100)
702     tree.column("Time", width=150)
703
704     v_scrollbar = ttk.Scrollbar(tree_frame, orient="vertical", command=tree.yview)
705     h_scrollbar = ttk.Scrollbar(tree_frame, orient="horizontal", command=tree.xview)
706     tree.configure(yscrollcommand=v_scrollbar.set, xscrollcommand=h_scrollbar.set)
707
708     tree.grid(row=0, column=0, sticky="nsew")
709     v_scrollbar.grid(row=0, column=1, sticky="ns")
710     h_scrollbar.grid(row=1, column=0, sticky="ew")
711
712     tree_frame.grid_rowconfigure(0, weight=1)
713     tree_frame.grid_columnconfigure(0, weight=1)
714
715     logs = get_all_logs()
716     for log in logs:
717         tree.insert("", tk.END, values=log)
718
719     tk.Label(content, text=f"Total Logs: {len(logs)}", fg="#888888",
720             bg=BG_MAIN, font=("Segoe UI", 10)).pack(pady=5)
721

```

Explanation: This function displays all logs in a table format:

- Clears the content area and adds a title.
- Creates a Treeview widget with columns for Username, Action, Risk, and Time.
- Sets column headings and widths for proper display.
- Inserts each log as a row in the Treeview.

4.9.8 Search Logs GUI

```
721
722 def search_logs_gui():
723     """Search logs by keyword in GUI."""
724     clear_frame(content)
725
726     tk.Label(content, text="SEARCH LOGS", fg=ACCENT, bg=BG_MAIN,
727             font=("Segoe UI", 18, "bold")).pack(pady=18)
728
729     # Search frame
730     search_frame = tk.Frame(content, bg=BG_MAIN)
731     search_frame.pack(pady=28)
732
733     tk.Label(search_frame, text="Enter Username or Action:", fg=TEXT,
734            bg=BG_MAIN, font=("Segoe UI", 11)).pack(side="left", padx=5)
735
736     search_entry = tk.Entry(search_frame, bg=HOVER, fg=TEXT,
737            font=("Segoe UI", 11), width=38)
738     search_entry.pack(side="left", padx=5)
739
740     # Results frame
741     results_frame = tk.Frame(content, bg=SIDEBAR_BG)
742     results_frame.pack(fill="both", expand=True, padx=28, pady=18)
743
744     # Treeview for results
745     tree = ttk.Treeview(results_frame, columns=("User", "Action", "Risk", "Time"),
746            show="headings")
747
748     tree.heading("User", text="Username")
749     tree.heading("Action", text="Action")
750     tree.heading("Risk", text="Risk Level")
751     tree.heading("Time", text="Timestamp")
752
753     tree.column("User", width=150)
754     tree.column("Action", width=400)
755     tree.column("Risk", width=100)
756     tree.column("Time", width=150)
757
758     scrollbar = ttk.Scrollbar(results_frame, orient="vertical", command=tree.yview)
759     tree.configure(yscrollcommand=scrollbar.set)
760
761     tree.pack(side="left", fill="both", expand=True)
762     scrollbar.pack(side="right", fill="y")
763
764     def perform_search():
765         keyword = search_entry.get().strip()
766         if not keyword:
767             messagebox.showwarning("Warning", "Please enter a search term")
768             return
769
770         # Clear previous results
771         for item in tree.get_children():
772             tree.delete(item)
773
774         # Call database search function
775         results = search_logs_db(keyword)
776         for result in results:
777             tree.insert("", tk.END, values=result)
778
779         status_label.config(text=f"Found {len(results)} result(s)")
780
781     # Search Button
782     tk.Button(search_frame, text="🔍 Search", bg=ACCENT, fg=TEXT,
783            font=("Segoe UI", 11, "bold"), command=perform_search).pack(side="left", padx=5)
784
785     # Status label
786     status_label = tk.Label(content, text="", fg="#888888",
787            bg=BG_MAIN, font=("Segoe UI", 18))
788     status_label.pack(pady=5)
```

Explanation: This function creates the search interface:

- Adds a title and search input field with label.
- Creates a Treeview widget to display search results.
- Defines a nested perform_search() function

4.9.9 Upload CSV

```
789
790 def upload_csv():
791     """Upload and process CSV file."""
792     filepath = filedialog.askopenfilename(
793         title="Select CSV File",
794         filetypes=[("CSV Files", "*.csv"), ("All Files", "*.*")]
795     )
796
797     if not filepath:
798         return
799
800     valid, msg = validate_csv(filepath)
801     if not valid:
802         messagebox.showerror("Validation Error", msg)
803         return
804
805     success, msg, count = import_csv(filepath)
806
807     if success:
808         messagebox.showinfo("Success", msg)
809         add_log(current_user, f"Uploaded CSV file with {count} logs", "Low")
810     else:
811         messagebox.showerror("Error", msg)
812
813 def export_logs():
814     """Export logs to CSV file."""
815     filepath = filedialog.asksaveasfilename(
816         title="Save CSV File",
817         defaultextension=".csv",
818         filetypes=[("CSV Files", "*.csv"), ("All Files", "*.*")]
819     )
820
821     if not filepath:
822         return
823
824     logs = get_all_logs()
825
826     if not logs:
827         messagebox.showwarning("Warning", "No logs to export")
828         return
829
830     success, msg = export_csv(filepath, logs)
831
832     if success:
833         messagebox.showinfo("Success", msg)
834         add_log(current_user, f"Exported {len(logs)} logs to CSV", "Low")
835     else:
836         messagebox.showerror("Error", msg)
```

Explanation: This function handles CSV file upload:

- Opens a file dialog for the user to select a CSV file.
- Returns if no file is selected (user cancelled).
- Shows error if validation fails.
- Imports the CSV using `import_csv()`.
- Shows error message on failure.

4.9.10 Export Logs

```
812
813 def export_logs():
814     """Export logs to CSV file."""
815     filepath = filedialog.asksaveasfilename(
816         title="Save CSV File",
817         defaultextension=".csv",
818         filetypes=[("CSV Files", "*.csv"), ("All Files", "*.*")]
819     )
820
821     if not filepath:
822         return
823
824     logs = get_all_logs()
825
826     if not logs:
827         messagebox.showwarning("Warning", "No logs to export")
828         return
829
830     success, msg = export_csv(filepath, logs)
831
832     if success:
833         messagebox.showinfo("Success", msg)
834         add_log(current_user, f"Exported {len(logs)} logs to CSV", "Low")
835     else:
836         messagebox.showerror("Error", msg)
```

Explanation: This function handles log export to CSV:

- Opens a save file dialog for the user to choose save location.
- Returns if no location is selected (user cancelled).
- Retrieves all logs using `get_all_logs()`.
- Shows warning if no logs exist to export.

4.9.11 Clear Logs

```
838 def clear_logs():
839     """Clear all logs after confirmation."""
840     if messagebox.askyesno("Confirm", "Are you sure you want to clear all logs?"):
841         if clear_all_logs():
842             messagebox.showinfo("Success", "All logs cleared successfully")
843             add_log(current_user, "Cleared all logs", "Medium")
844         else:
845             messagebox.showerror("Error", "Failed to clear logs")
846
```

Explanation: This function clears all logs with confirmation:

- Shows a confirmation dialog using askyesno().
- Proceeds only if user confirms.
- Calls clear_all_logs() to delete all logs from database.
- Shows success message and logs the action if successful.
- Shows error message if database operation fails.

4.9.12 Bar Graph

```

846
847 def show_bar_graph():
848     """Display bar graph of risk distribution."""
849     stats = get_risk_statistics()
850
851     if stats['total'] == 0:
852         messagebox.showwarning("Warning", "No data to display")
853         return
854
855     plt.figure(figsize=(10, 6))
856
857     risks = ['High', 'Medium', 'Low']
858     counts = [stats['high'], stats['medium'], stats['low']]
859     colors = [HIGH, MEDIUM, LOW]
860
861     bars = plt.bar(risks, counts, color=colors, edgecolor='white', linewidth=2)
862
863     plt.title('Risk Distribution Analysis', fontsize=16, fontweight='bold', pad=20)
864     plt.xlabel('Risk Level', fontsize=12)
865     plt.ylabel('Number of Incidents', fontsize=12)
866
867     for bar, count in zip(bars, counts):
868         height = bar.get_height()
869         plt.text(bar.get_x() + bar.get_width()/2., height,
870                 f'{count}', ha='center', va='bottom', fontsize=11, fontweight='bold')
871
872     plt.grid(axis='y', alpha=0.3)
873     plt.tight_layout()
874     plt.show()
875

```

Explanation: This function creates a bar graph of risk distribution:

- Retrieves risk statistics using get_risk_statistics().
- Shows warning if no data exists.
- Creates a matplotlib figure with specified size.
- Defines risk categories, counts, and colors.
- Creates bar chart with color-coded bars.

4.9.13 Pie Chart

```
876 def show_pie_chart():
877     """Display pie chart of risk percentages."""
878     stats = get_risk_statistics()
879
880     if stats['total'] == 0:
881         messagebox.showwarning("Warning", "No data to display")
882         return
883
884     plt.figure(figsize=(8, 8))
885
886     labels = ['High Risk', 'Medium Risk', 'Low Risk']
887     sizes = [stats['high'], stats['medium'], stats['low']]
888     colors = [HIGH, MEDIUM, LOW]
889
890     non_zero = [(l, s, c) for l, s, c in zip(labels, sizes, colors) if s > 0]
891     if non_zero:
892         labels, sizes, colors = zip(*non_zero)
893
894     wedges, texts, autotexts = plt.pie(sizes, labels=labels, colors=colors,
895                                     autopct='%1.1f%%', startangle=90,
896                                     textprops={'fontsize': 12, 'fontweight': 'bold'})
897
898     plt.title('Risk Distribution Percentage', fontsize=16, fontweight='bold', pad=20)
899     plt.axis('equal')
900     plt.tight_layout()
901     plt.show()
```

Explanation: This function creates a pie chart of risk percentages:

- Creates pie chart with percentage labels.

4.9.14 Logout

```
903 def logout():
904     """Handle logout."""
905     if messagebox.askyesno("Logout", "Are you sure you want to logout?"):
906         global current_user, current_role
907
908         add_log(current_user, "Logged out of system", "Low")
909
910         current_user = None
911         current_role = None
912
913         create_login_screen()
914
```

Explanation: This function handles user logout:

- Shows confirmation dialog using askyesno().
- Proceeds only if user confirms.
- Logs the logout action to the database.
- Clears session variables (current_user, current_role).
- Returns to the login screen using create_login_screen().

4.9.15 On Closing

```

903 def logout():
904     """Handle logout."""
905     if messagebox.askyesno("Logout", "Are you sure you want to logout?"):
906         global current_user, current_role
907
908         add_log(current_user, "Logged out of system", "Low")
909
910         current_user = None
911         current_role = None
912
913         create_login_screen()
914
915 def on_closing():
916     """Handle application closing."""
917     if messagebox.askokcancel("Quit", "Do you want to quit the application?"):
918         if conn:
919             conn.close()
920         root.destroy()

```

Explanation: This function handles application shutdown:

- Shows confirmation dialog when user closes the window.
- Closes the database connection if it exists.
- Destroys the main window to exit the application.

4.10 Main Function

```
app.py > ...
922 # ===== MAIN FUNCTION =====
923 def main():
924     """Main entry point for the application."""
925     global root
926
927     try:
928         init_database()
929
930         root = tk.Tk()
931         root.title("Insider Threat Detection System")
932         root.geometry("1200x750")
933         root.configure(bg=BG_MAIN)
934
935         root.protocol("WM_DELETE_WINDOW", on_closing)
936
937         create_login_screen()
938
939         root.mainloop()
940
941     except Exception as e:
942         print(f"Fatal error: {e}")
943         messagebox.showerror("Fatal Error", f"Application failed to start: {e}")
944
945 if __name__ == "__main__":
946     main()
```

Explanation: This is the main entry point of the application:

- Initializes the database using `init_database()`.
- Creates the main tkinter window with title and size.
- Sets the window background color.
- Configures the close handler to call `on_closing()`.

This completes the implementation chapter, covering all major components of the Insider Threat Detection System including custom data structures, database functions, risk classification, CSV processing, activity simulation, and the complete GUI interface.

5.0 RESULTS AND ANALYSIS

The Insider Threat Detection System was tested across all functional modules and produced successful results. The login system correctly authenticated valid credentials (admin/admin123) while rejecting invalid attempts with appropriate error messages. The risk classification algorithm accurately categorized test actions, with "Deleted confidential file" returning High risk, "Downloaded customer data" returning Medium risk, and "Logged into system" returning Low risk, demonstrating effective keyword-based threat assessment. The custom stack implementation correctly followed LIFO order, with push and pop operations returning the expected items, while the circular queue maintained FIFO order even when wrapping around the buffer. Database operations functioned properly, with user verification, log addition, and search queries all returning correct results. CSV import successfully processed valid files while rejecting malformed formats, and export functionality created properly formatted CSV files.

5.1 CONCLUSION AND RECOMMENDATION

5.1.1 Conclusion

This coursework successfully developed an Insider Threat Detection System using Python. The system monitors user activities and classifies them into High, Medium, and Low risk levels.

What I Accomplished:

- Implemented secure authentication with SHA-256 password hashing
- Created custom stack and queue data structures
- Developed a risk classification algorithm using keyword matching
- Built a SQLite database for storing users and logs
- Added CSV import/export functionality
- Created data visualizations using matplotlib
- Designed a user-friendly GUI with dark theme

Key Learnings:

- Understanding of data structures (stack, queue, circular buffer)
- Database integration with SQLite
- GUI development with tkinter
- File handling with CSV
- Password security with hashing algorithms

Challenges Faced:

- Implementing circular queue logic with modulo arithmetic
- Managing global variables across multiple functions
- Handling database errors gracefully

5.2 Recommendations for Future Work

The following improvements can be made to enhance the system:

- Machine Learning Integration: Implement anomaly detection algorithms to identify unusual patterns automatically
- Real-time Monitoring: Add socket programming for live activity feeds with immediate alerts
- Multi-factor Authentication: Add email or SMS verification for enhanced security
- Role-based Access: Implement multiple user roles with different permission levels
- Cloud Database: Migrate to MySQL or PostgreSQL for networked deployments
- Mobile Interface: Create a mobile app for remote monitoring

These enhancements would make the system more robust and suitable for enterprise environments.

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